Introduction to Computer Programming
Fall 2026
Midterm

First Name	
Last Name	
NetID	

Instructions

- **Duration**: 9:50 AM-11:30 AM (100 minutes)
- **Max score**: 100 pts

- Computers, calculators, phones, textbooks or notebooks are **not allowed**
- When writing programs, make very clear your line **indentations** (there are some indentation levels in the areas where you will write code). If indentation is not perfectly clear, the grader will consider "no indent".
- We can **not answer** questions during the test.
- If you are unsure of a problem, please write down your assumptions, and proceed with it. We will grade accordingly.
- **Do not** use lists, string methods (such as .split() or .count()), dictionary, set or define functions. Use only concepts covered in class.

Grading

Questions	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**
Points	$8'$	$8'$	$24'$	$10'$	$10'$	$10'$	$15'$	$15'$
Score								
Total								

Question 1 | Boolean Expression

Evaluate the following boolean expression:

`'ace' in 'space' or 5%2 < 1`	True
`3 == '3' or 'pen' > 'pencil'`	False
`7//3 > 2 or len('abs') == 3 or not '12' == 12`	True
`not True and True and True`	False

Question 2 | Binary and Hexadecimal Representaion

Complete the table below by converting each given number into the other two number systems: decimal, binary, and hexadecimal.

Decimal	Binary	Hexadecimal
43	0b101011	0x2b
90	0b1011010	0x5A

Question 3 | Predict the Output

Program 1	Output 1

`print( 21 // 5 )`
`print( 21 % 5 )`
`print( f'{21/5:.2f}' )`
	4
	1
	4.20

Program 2	**Output 2**

`a = 2`
`b = 5`
`print(a , b, sep='*', end = '=')`
`print(a * b)`
`print('Done!')`
	2*5=10
	Done!

Program 3	**Output 3**

`word = "PROGRAM"`
 	
`print(word[3])`
`print(word[-4:len(word)]`
`print(word[:3])`
`print(word[1:-1:2])`
`print(word[-2::-2])`
	G
	GRAM
	PRO
	RGA
	AGR

Program 4	Output 4

`count = 0`
 	
`for i in range(1, 13):`
`  if count > 4:`
`  break`
 	
`  if i % 2 == 0 or i % 3 == 0:`
`  print(i)`
`  count += 1`
 	
`print("End")`
	2
	3
	4
	6
	8
	End

Program 5	**Output 5**

`res, n = 1, 1`
 	
`while n < 5:`
`  res *= n`
`  n += 1`
 	
`print(res)`
	24

Program 6	**Output 6**

`result = ''`
 	
`for ch in 'bAeBe3':`
`  if 'a' < ch < 'z':`
`  result += ch`
 	
`print(result)`
	bee

Program 7	**Output 7**

`S = '1234'`
 	
`if len(S) > 3:`
`  print('One')`
`elif int(S) % 2 == 0:`
`  print("Two")`
`else:`
`  print("Three")`
	One

Program 8	**Output 8**

`total = 0`
 	
`for i in range(7, 1, -2):`
`  total += i`
 	
`print(total)`
	15

Question 4 | Predict the Output

```
N = aString

i, j = 0, 0
a = 0

while i < len(N):

    j = i + 1
    while j < len(N) and N[j] == N[i]:
        j = j + 1

    if j != i + 1:
        a = a + 1

    i = j

print(a)
```

Fill in the table with the expected output for each value of `aString`.

`aString`	Expected output
`'abc'`	0
`'aa'`	1
`'aabbcc'`	3
`'aaaabccdeeffffgg'`	5

Question 5 | Character Classifier

Write a Python program that reads a single character and classifies it as follows:

- If it is an English letter:
  - Display whether it is ***uppercase*** or ***lowercase***.
  - Display whether it is a ***vowel*** `(a,e,i,o and u)` or a ***consonant***.
- If it is a digit from `0` through `9`, display whether it is ***even*** or ***odd***.
- Otherwise, print ***Other character***.

Hint: *Use 'A' <= letter <= 'Z' to check whether letter is an uppercase letters; similarly for lowercase letters and digits*

Example Output (user input is in bold)**:**

Example 1:

```
Enter a character: E
Uppercase vowel
```

Example 2:

```
Enter a character: b
Lowercase consonant
```

Example 3:

```
Enter a character: 7
Odd digit
```

Example 4:

```
Enter a character: #
Other character
```

Write your code on the next page!

Your code:

```
letter = input("Enter a single character: ")

if 'A' <= letter <= 'Z' or 'a' <= letter <= 'z':
    if 'A' <= letter <= 'Z':
        print("Uppercase")
    else:
        print("Lowercase")

    if letter in "aeiouAEIOU":
        print("Vowel")
    else:
        print("Consonant")

elif '0' <= letter <= '9':
    if int(letter) % 2 == 0:
        print("Even")
    else:
        print("Odd")

else:
    print("Other character")
```

Question 6 | Saving Goal

Write a program that asks for an initial deposit, an annual interest rate (in %), and a target amount. Each year the balance grows by the interest rate.

The balance of next year can be calculated as:

$$balance * (1 + rate/100)$$

Print the balance at the end of each year (2 decimal places) until the balance reaches or exceeds the target, then print how many years were needed.

Example Output (user input is in bold)**:**

Example 1:

```
Initial deposit: 1000
Annual interest rate (%): 4
Target amount: 1200
Year 1: 1040.00
Year 2: 1081.60
Year 3: 1124.86
Year 4: 1169.86
Year 5: 1216.65
Years needed: 5
```

Write you code on the next page!

Your code:

```
balance = float(input("Enter the initial deposit: "))
rate = float(input("Enter the annual interest rate (%): "))
target = float(input("Enter the target amount: "))

years = 0

while balance < target:
    balance = balance * (1 + rate / 100)
    years += 1
    print(f"Year {years}: {balance:.2f}")

print(f"Years needed: {years}")
```

Question 7 | Vending Machine

A vending machine only accepts `1, 5, 10` and `20` yuan bills/coins. Write a Python program that:

- asks for the item price (you may assume it is a positive integer, in yuan);
- repeatedly asks the user to insert money until the total paid is at least the price. Reject any value that is not `1, 5, 10` or `20` by printing ***Rejected: <value>***. After each accepted payment, print the progress;
- returns the change using **as few `10, 5` and `1` yuan pieces as possible**, printing only the denominations that are actually used. If there is no change, print ***No change. Enjoy your drink!***

Example Output (user input is in bold)**:**

Example 1:

Item price (yuan): **7**
Insert money (1, 5, 10 or 20): **3**
Rejected: 3
Insert money (1, 5, 10 or 20): **20**
Paid: 20 / 7
Change: 13
10 yuan x 1
1 yuan x 3
Enjoy your drink!

Example 2:

Item price (yuan): **8**
Insert money (1, 5, 10 or 20): **5**
Paid: 5 / 8
Insert money (1, 5, 10 or 20): **2**
Rejected: 2
Insert money (1, 5, 10 or 20): **1**
Paid: 6 / 8
Insert money (1, 5, 10 or 20): **1**
Paid: 7 / 8
Insert money (1, 5, 10 or 20): **1**
Paid: 8 / 8
No change. Enjoy your drink!

Write you code on the next page!

Your code:

```
price = int(input("Enter the item price: "))
paid = 0

while paid < price:
    money = int(input("Insert money (1, 5, 10 or 20): "))

    if money == 1 or money == 5 or money == 10 or money == 20:
        paid += money
        print(f"Paid: {paid} yuan / Price: {price} yuan")
    else:
        print(f"Rejected: {money}.")

change = paid - price

if change == 0:
    print("No change. Enjoy your drink!")
else:
    tens = change // 10
    change = change % 10
    fives = change // 5
    ones = change % 5

    if tens > 0:
        print(f"10 yuan: {tens}")
    if fives > 0:
        print(f"5 yuan: {fives}")
    if ones > 0:
        print(f"1 yuan: {ones}")
```

Question 8 | Prime Finder

Write a Python program that reads two integers **`start`** and **`end`** (you may assume `start ≤ end`) and prints all prime numbers in that range (inclusive) on one line, separated by spaces, followed by the total count.

A prime is an integer greater than `1` whose only divisors are `1` and itself.

Example Output (user input is in bold)**:**

Exampe 1:

Start: **10**
End: **30**
11 13 17 19 23 29
Total primes: 6

Example 2:

Start: **1**
End: **10**
2 3 5 7
Total primes: 4

Your code:

```
start = int(input("Enter start: "))
end = int(input("Enter end: "))
count = 0

for number in range(start, end + 1):
    if number > 1:
        is_prime = True

        for divisor in range(2, int(number ** 0.5) + 1):
            if number % divisor == 0:
                is_prime = False
                break

        if is_prime:
            print(number, end=" ")
            count += 1

print()
print(f"Total prime: {count}")
```